Medibotx: An Autonomous Delivery And Navigation System For Healthcare

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Abstract - MediBotX is an innovative project developed by mechanical engineering undergraduates to address the evolving needs of the healthcare sector. This ground-breaking initiative aims to revolutionize patient care by designing and building a small, autonomous robot capable of seamlessly delivering essential items, including medicines and food, within hospital premises. Through meticulous research and a focus on optimizing patient care, our team has developed a prototype showcasing promising capabilities in navigating hospital environments, obstacle avoidance, and secure item delivery.

Keywords – Autonomous Delivery, Navigation System, Healthcare Robotics, Patient Care, Hospital Efficiency.

I. INTRODUCTION

In the rapidly evolving landscape of healthcare, the convergence of technology and patient care has become increasingly prominent. Against the backdrop of escalating healthcare costs, resource constraints, and the imperative to enhance patient experiences, there exists a pressing need for innovative solutions to optimize healthcare logistics. In response to these challenges, our project, "MediBotX: An Autonomous Delivery and Navigation System for Healthcare," emerges as a pioneering endeavor aimed at reshaping the delivery of essential items within healthcare facilities. With a focus on enhancing efficiency, accuracy, and patient-centric care, MediBotX represents a paradigm shift in the integration of autonomous robotics into healthcare delivery systems. By harnessing cutting-edge technologies such as line following navigation, RFID technology, and the ESP32 microcontroller, MediBotX seeks to streamline delivery processes, mitigate patient care delays, and optimize resource utilization. Moreover, the project aspires to contribute to the broader discourse on healthcare innovation by addressing critical challenges and introducing scalable solutions that align with the evolving needs of healthcare facilities. Through meticulous research, technological innovation, and real-world testing, MediBotX aims to redefine the contours of healthcare logistics, guided by principles of efficiency, innovation, and an unwavering commitment to advancing patient well-being.

Keywords – Healthcare Innovation, Autonomous Robotics, Patient Experience, Resource Optimization, Technological Advancements.

II. LITERATURE SURVEY

1. Applications of Line Follower Robot in Medical Field, Tushar Jain [2014]. This paper concludes that healthcare costs, especially in resource-constrained regions like India, are driven by expensive machinery and skilled labor. It suggests that autonomous robotic systems, like MediBotX, could replace human labor, improving patient care efficiency, accuracy, and reducing per capita healthcare expenses.

2. Design and Fabrication of Medicine Delivery Robots for Hospitals, Dr. K. Lakshmi Narayanan [2019]. The research paper highlights the preference for the Atmega 2560 board due to its popularity in robot building. The Nursing Robot ensures timely medicine delivery, utilizing IR sensors, RFID room detection, and obstacle detection. It employs DC motors for four-directional movement, utilizes IoT technology, and eliminates the need for human intervention in patient medication reminders.

3. A Line Follower Robot from design to Implementation: Technical issues and problems, M. Mehdi Sanaatiyan [2010]. The research paper concludes that their current line-following robot is highly capable, able to navigate various curves and cycles efficiently. They emphasize the importance of a lightweight design, high-speed motors, and sensitive sensors to optimize distance coverage and speed. The robot incorporates eight infrared sensors, an ATMega16 microcontroller, and an L298 driver to control motor direction and speed.

4. Performance Analysis of Scheduling Multiple Robots for Hospital Logistics, Seohyun Jeon and Jaeyeon Lee [2017]. The research paper explores autonomous mobile robots for hospital logistics, presenting task allocation algorithms. Through simulations, it finds that MT-COM performs better with larger fleets, offering recommendations for fleet size optimization. Future work will address the task binding problem (MTTA).

5. Line Following Robot Using Arduino for Hospitals, Jagruti Chaudhari [2019]. The research paper highlights the use of a 9V battery for DC motor operation in a prototype robot designed for hospital applications. It emphasizes the need for improved power supply and component upgrades to handle higher loads effectively. The paper anticipates that with better components and adjustments in technical aspects, the robot could become a valuable asset for healthcare centers, especially in transporting high-load goods and enhancing efficiency.

6. Robot for Delivery of Medicines to Patients Using Artificial Intelligence in Health Care, Srikanth Kavirayani [2013]. The research paper highlights limitations in assuming static ward arrangements and precise patient location data, making dynamic updates and the presence of other robots challenging. It underscores the complexity of autonomous delivery systems in a constantly evolving healthcare landscape.

III. METHODOLOGY

(1) Problem Definition:

The problem addressed in the research paper is the efficient delivery of essential items within healthcare facilities. Manual systems often cause delays, inefficiencies, and inconsistent service, impacting patient care. These challenges are exacerbated in resource-constrained regions like India and during the COVID-19 pandemic, which emphasizes the need for contactless delivery. "MediBotX: An Autonomous Delivery and Navigation System for Healthcare" aims to develop an autonomous robot for precise and reliable item delivery, enhancing efficiency, reducing costs, and improving patient care through safe, contactless methods.

(2) Problem planning and Designing the Robot:

2.1: Design conceptualization of robot: The robot design involves the creation and assembly of a sturdy and highly maneuverable hardware chassis. For our prototype, we have opted to use a plywood and acrylic sheet platform. which provides solid foundation for the robot structure. please refer to figure1 below for an illustration of the robot’s design with the plywood and acrylic sheet platform.

2.2 To facilitate the autonomy of the bot, we will utilize line-following, RFID, object detection, and the WiFi module of the ESP32 microcontroller. We have chosen the ESP32 for its integrated WiFi module, allowing seamless connectivity between the robot and user (patient) Android devices. By leveraging this pre-existing technology, we can easily control the robot's path without needing to develop a separate custom app. This setup provides a convenient and efficient solution for autonomously operating the robot using the ESP32 microcontroller.

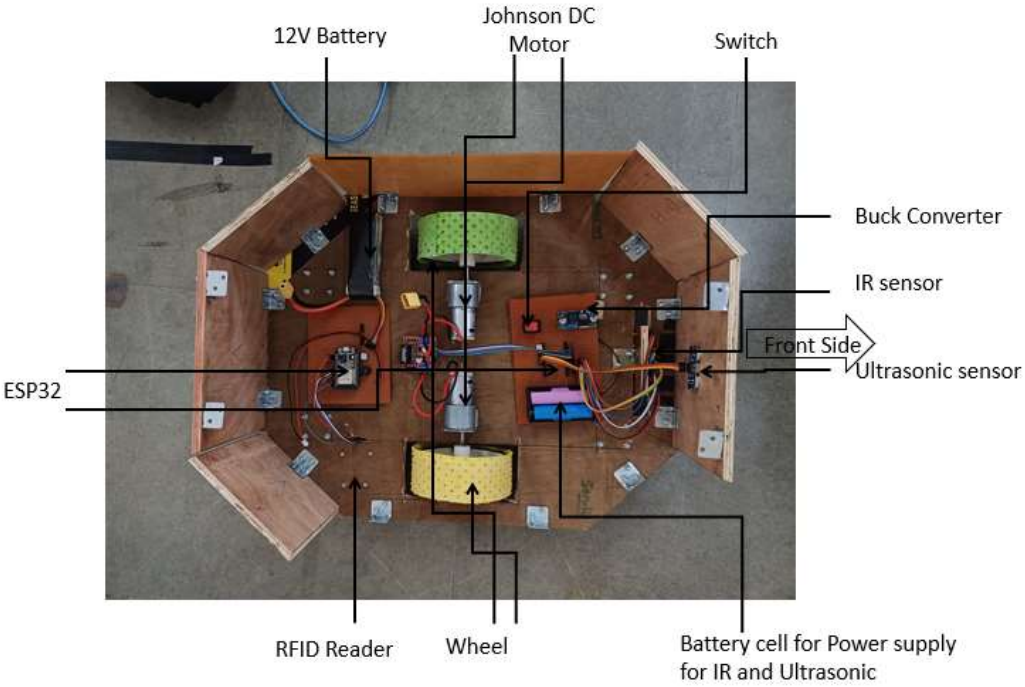


Figure 1: Robot's Schematic Diagram

2.3 Implement Line Follower system: The line follower mechanism is a key part of MediBotX autonomous navigation. Using an array of IR sensors, it detects and follows a predefined path marked on the hospital floor. The sensors provide real-time feedback to the ESP32 microcontroller, which adjusts the robot's movements to stay on course.

2.4 Implement Obstacle avoidance system: The obstacle avoidance system uses an ultrasonic sensor mounted at the front of the robot to detect objects or humans in its path. When an obstacle is detected, the robot stops and waits until the path is clear, ensuring safety for both the robot and its surroundings.

2.5 Implement RFID tag and reader system: The RFID tag and reader system allows the robot to identify specific patients or destinations. When a patient calls the bot using a website interface, the robot navigates to the patient's location and stops when it detects the corresponding RFID tag, ensuring precise delivery.

2.6 Implement WIFI module system: The WIFI module integrated into the ESP32 microcontroller enables wireless connectivity between the robot and the hospital's network. This allows patients to call the robot via a web interface, enhancing user convenience and interaction

2.7 Implement Electric power supply: The robot is powered by a 12V battery that supplies electricity to all its components, including the motors, sensors, and microcontroller. The power system is designed to ensure reliable and continuous operation throughout its tasks.

2.8 Components Used:

Table 1. Components Used with Voltage

Components (Sensor/ actuator)	
Sensor/ Actuator	Working Voltage
Johnon DC Motor	12V
L298 Motor Driver	12V
ESP32	5V
IR sensor	5V
Ultrasonic sensor	5V
RFID Tag and Reader	3.3V
Buzzer	5V
Voltage sensor	5V
Buck Converter	-
Jumper Wires	-

(3) Working:

3.1 The implementation of an autonomous delivery and navigation system for healthcare can be deployed in various healthcare settings such as hospitals, clinics, or assisted living facilities. Within these environments, the robotic system can navigate through corridors, hallways, and designated pathways to deliver medication, medical supplies, documents, or other essential items to different departments, patient rooms, or healthcare professionals.

3.2 By integrating seamlessly into existing infrastructure, the system enhances operational efficiency, reduces manual labor, and minimizes the risk of errors in healthcare logistics.

3.3 Additionally, the robot's autonomous capabilities ensure safe and reliable navigation even in busy healthcare environments, contributing to improved patient care and staff productivity.

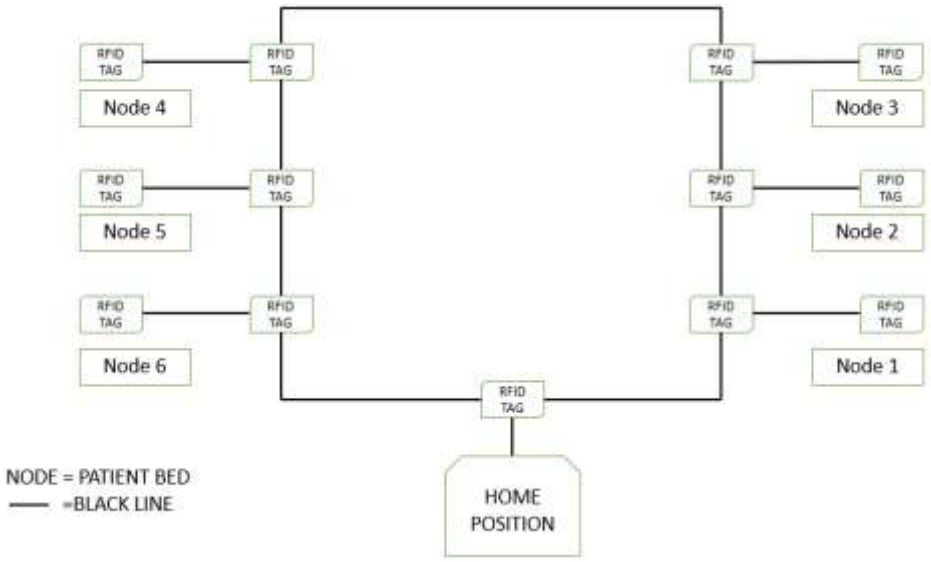


Figure 2: Map for Navigation

3.4 Our bot utilizes line following for navigation, with the ESP32 microcontroller at its core, featuring an integrated WIFI module for seamless connection to the hospital's WIFI system.

3.5 When a patient needs assistance, they simply scan the QR code beside them, redirecting them to a webpage (refer figure 3) where they can call the bot by selecting their bed number.

3.6 For example, a patient in bed number 4 would scan the QR code and press '4' on the webpage. Upon activation, the bot travels to the designated bed, triggering a buzzer upon arrival to alert the patient. Additionally, the bot is equipped with a switch for manual operation when necessary.

3.7 After completing the task for patient number 4, the patient must press the switch to signal completion; otherwise, the bot automatically returns to its home position after 2 minutes.

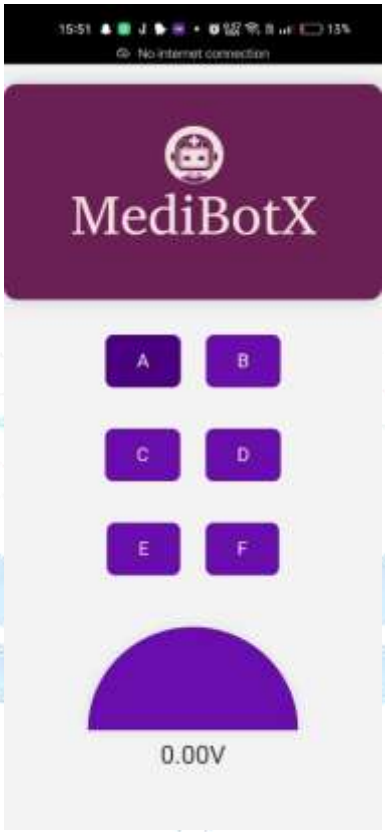


Figure 3: Webserver Interface

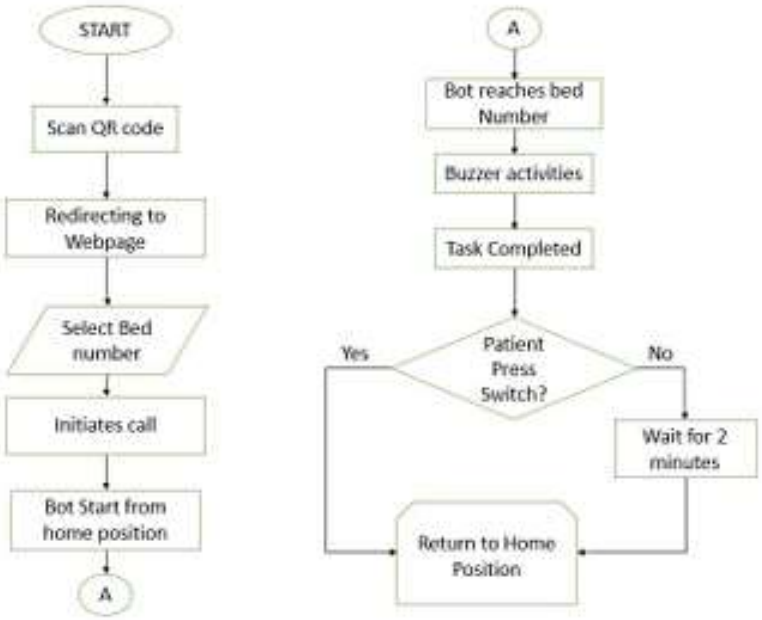
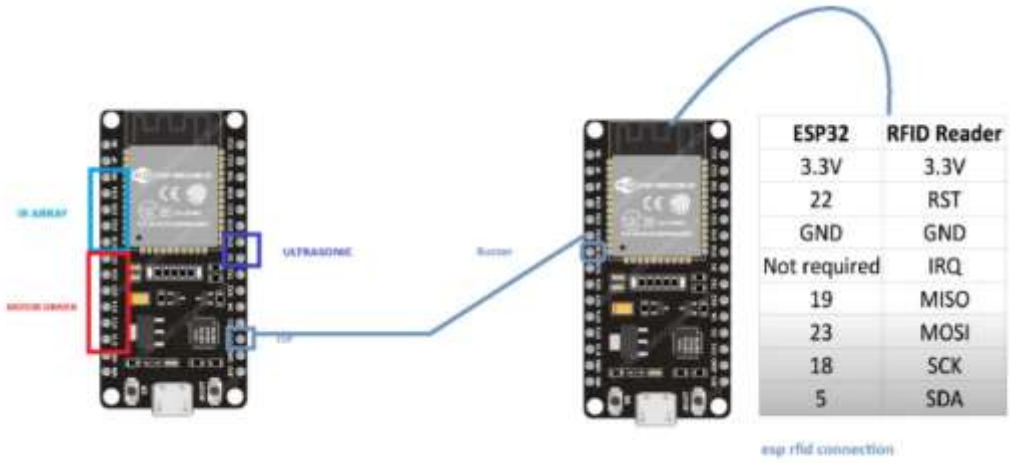


Figure 4: Flowchart



ESP32	RFID Reader
3.3V	3.3V
22	RST
GND	GND
Not required	IRQ
19	MISO
23	MOSI
18	SCK
5	SDA

Figure 5: Circuit Diagram

IV. RESULTS

1. The bot uses five IR sensors (IR1, IR2, IR3, IR4, and IR5) positioned from left to right to detect a black line on the floor. Each sensor outputs a value of 1 when it detects the line and 0 when it does not. The navigation logic is as follows: If IR2 outputs 1 while all other sensors output 0, the bot moves forward. If IR1 or IR2 outputs 1 and all other sensors output 0, the right motor rotates, causing the bot to turn left. Conversely, if IR4 or IR5 outputs 1 and all other sensors output 0, the left motor rotates, causing the bot to turn right. If all sensors output 0, the bot rotates on its own axis to search for the line.

2. We are using one ultrasonic sensor in front of the robot. If any obstacle or human comes in front of the robot, it stops and remains stationary until the obstacle is removed from its path. This feature ensures the safety of the robot, infrastructure, and humans. Once the obstacle is removed, the robot continues on its path until it reaches its destination.

3. We are using RFID tags and a reader to detect patients or destinations. When the bot has no tasks, it stays at the home position. When a patient calls the bot using the website interface (refer to Fig. 3), the bot starts from the home position and moves to the designated destination (patient number). Upon detecting the patient's RFID tag, the bot stops and activates the buzzer. After reaching the desired patient, the bot stays for 10 seconds and then returns to its home position.

4. When the bot is returning to the home position and another patient calls it, the bot will go directly to the new patient's location instead. This shows the bot's flexibility. It helps the bot respond quickly to patient requests, reduces idle time, and improves service efficiency.

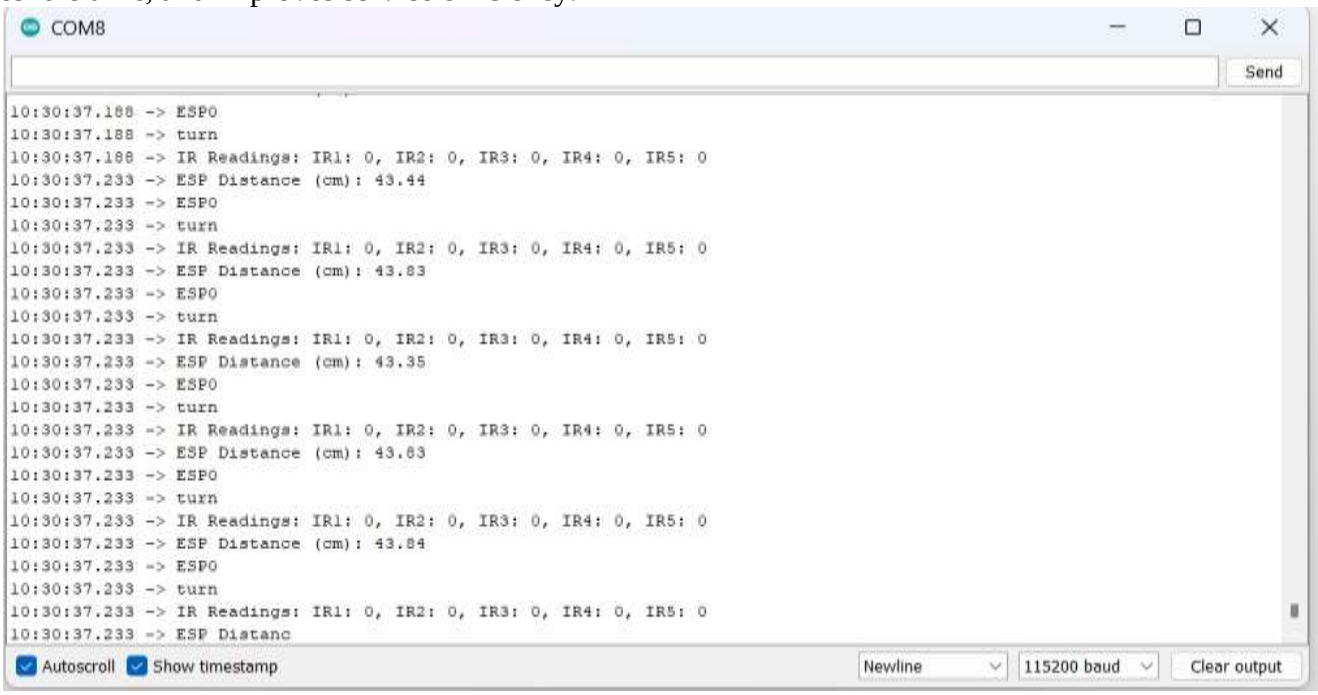


Figure 6: Output screen on Arduino IDE

V. FUTURE SCOPE

1. Implementing PID on Line follower: Using a PID (Proportional-Integral-Derivative) controller can improve the precision and stability of the line-following mechanism, allowing the robot to navigate more smoothly and accurately.
2. Implementation of Swarm Robotics: Introducing swarm robotics can enable multiple MediBotX units to work together, coordinating tasks and improving efficiency in delivering items within large hospital environments.
3. Implementation of Robotics arm on the Robot: Adding a robotic arm can expand the bot's functionality to include tasks like picking up and placing items, further aiding healthcare staff and enhancing patient care.
4. Implementation of machine Learning algorithm: Incorporating machine learning algorithms can help MediBotX adapt to different environments and learn from its experiences, improving its performance over time.
5. Implementation for Multi-Tasking: Enhancing the bot's capability to handle multiple tasks simultaneously will increase its efficiency, allowing it to perform various duties such as delivering meals, medications, and other supplies.
6. Implementation of Personal Assistance that can guide Patient: Equipping the bot with personal assistance features can help guide patients within the hospital, providing directions and escorting them to different locations.
7. Integration with IOT devices: Integrating IoT devices can enhance the bot's functionality by enabling real-time data exchange and monitoring, improving coordination and response times in healthcare settings.
8. Implementation of autonomous Battery Charging system: Developing an autonomous battery charging system will ensure that the bot remains operational with minimal human intervention, automatically returning to a charging station when needed.

VI. CONCLUSION

In conclusion the MediBotX project is a big step forward in using robots in healthcare. It uses line-following technology, RFID for identifying patients, and sensors to avoid obstacles, making it efficient and reliable for delivering medications and meals. The ESP32 microcontroller with WIFI ensures easy connectivity and patient interaction. This setup reduces the workload for healthcare staff and minimizes human contact, which is crucial during pandemics like COVID-19. Made from strong materials like plywood and acrylic sheets and equipped with multiple sensors, MediBotX is durable and effective. The project shows how technology can improve patient care and efficiency in hospitals.

VII. REFERENCES

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